CS & DA
Probability and Statistics



Probability

Lecture No. 02



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Recap of previous lecture



Topic

Probability (BASICS)

Topics to be Covered



Topic

BASICS of PROBABILITY (Part-02)

(*) Fav Cases :→ Fav = { which is Required & should be assumed as Fav }

(*) Revision : = { Desired outcome " " " " }

Operation	P ∈ C	Prob	ME	Ind
Either or	Add	∪ use Addition Th	$P(A \cup B) = P(A) + P(B)$	$P(A \cup B) = P(A) + P(B) - P(A) \cdot P(B)$
AND	Multiply	∩ use Multi Th	$P(A \cap B) = 0$	$P(A \cap B) = P(A) \cdot P(B)$

(*) (1) Three Dice are thrown simultaneously
(2) A Die is thrown three times

Both have same S space

$S = \{(111)(112)\ldots(116)\ (211)(212)\ldots(666)\} \Rightarrow n(S) = 6 \times 6 \times 6 = 216$ Triplets

HW8 7 surgical strikes occurred in a week from INDIA on PAKISTAN then find the prob that all will occur on a same day ?

Sol: App① : it is not easy to write sample space in terms of set ☹

App② : S.S = 7 Days = 7 so Total ways of occurring 5 Strikes = $\frac{7}{S_1}\times\frac{7}{S_2}\times\frac{7}{S_3}\times\frac{7}{S_4}\times\frac{7}{S_5}\times\frac{7}{S_6}\times\frac{7}{S_7}$ (No Restriction / RA)

$= 7^7$ ways.

fav ways of occurring SS = All will occur on (Same day)

$= \frac{7}{S_1}\times\frac{1}{S_2}\times\frac{1}{S_3}\times\frac{1}{S_4}\times\frac{1}{S_5}\times\frac{1}{S_6}\times\frac{1}{S_7} = 7$ ways

Hence Req Prob $= \frac{\text{fav}}{\text{Total}} = \frac{7}{7^7} = \frac{1}{7^6}$ eg (M M M M M M M M)

(*) No. of ways to select one day from 7 days $= {}^{7}C_1 = 7$ ways $=$ Say.

(*)

	S_1	S_2	S_3	S_4	S_5	S_6	S_7
Case I	(M	M	M	M	M	M	M)
" 2 or	(T	T	T	T	T	T	T)
" 3 or	(W	W	W	W	W	W	W)
" 4 or	(Th	Th	Th	Th	Th	Th	Th)
" 5 or	(F	F	F	F	F	F	F)
" 6 or	(Sat	Sat	Sat	Sat	Sat	Sat	Sat)
" 7 or	(S	S	S	S	S	S	S)

So far Cases $= 1+1+1+1+1+1+1 = 7$ ways

Q: Two Dices are thrown simultaneously then write it's S.space

$$S=\begin{Bmatrix}(1,1),(1,2),(1,3)\ldots(1,6),(2,1),(2,2)\ldots(2,6)\\(3,1),(3,2)\ldots(3,6),(4,1),(4,2)\ldots(4,6)\\(5,1),(5,2)\ldots(5,6),(6,1),(6,2)\ldots(6,6)\end{Bmatrix}\Rightarrow n(S)=\underset{D_1}{6}\times\underset{D_2}{6}=36 \text{ pair}$$

① Find the prob that sum of the outcomes on Dices will be 8 ?

App I $A=\{\text{sum is } 8\}=\{(2,6),(6,2),(3,5),(5,3),(4,4),(1,7),(7,1)\}\Rightarrow n(A)=5$

$$\text{So } P(A)=\frac{n(A)}{n(S)}=\frac{5}{36}$$

② App I Find the Prob that sum is 9 ?

$B=\{\text{sum is } 9\}=\{(6,3),(3,6),(5,4),(4,5)\}\Rightarrow n(B)=4$ So $P(B)=\frac{n(B)}{n(S)}=\frac{4}{36}$

$(4,5),(4,5)$

③ Find the Prob that Sum is Both 8 and 9 ?

App III ∵ $A \cap B = \phi$ is A & B are (ME) Events so $P(A \cap B) = 0$

④ Find the Prob that Sum is either 8 or 9 ?

App III ∵ A & B are (ME) so $P(A \cup B) = ? = P(A) + P(B) - 0$

$$= \frac{5}{36} + \frac{4}{36} = \frac{9}{36} = \frac{1}{4}$$

Gate

⑤ Find the Prob that Sum is Neither 8 Nor 9 ?

App II $P(\text{Sum is Neither 8 Nor 9}) = 1 - P(\text{Sum is either 8 or 9})$

$$P(\bar{A} \cap \bar{B}) = 1 - P(A \cup B) = 1 - \frac{1}{4} = \frac{3}{4}$$

Gate

⑥ Find the prob that all the outcomes are Identical ?

App I $C = \{$ Identical outcomes $\} = \{(11), (22), (33), (44), (55), (66)\} \Rightarrow n(C) = 6$

So $P(C) = \frac{6}{36}$

Gate

⑦ Find the prob that product of the outcomes will be a perfect square ?

App I $D = \{$ Product will be perfect sq $\} = \{(11), (22), (33), (44), (55), (66), (14), (41), \cancel{(94)}, \cancel{(49)}\}$

$n(D) = 8$ So $P(D) = \frac{n(D)}{n(S)} = \frac{2}{36}$

⑧ Find the prob that Sum exceeds 9 ?

App I: Do yourself ; Trick: $P(\text{Sum} > 9) = P(\text{Sum} = 10 \text{ or } 11 \text{ or } 12) = \frac{3+2+1}{36} = \frac{6}{36}$

(9) Find the prob that sum will be a prime No?

Abb I → Do yourself, Trick: $P(\text{sum} = \text{prime No}) = P(\text{sum} = 2 \text{ or } 3 \text{ or } 5 \text{ or } 7 \text{ or } 11)$

$= \frac{1+2+4+6+2}{36} = \frac{15}{36}$

(10) find the prob that sum is divisible by 4?

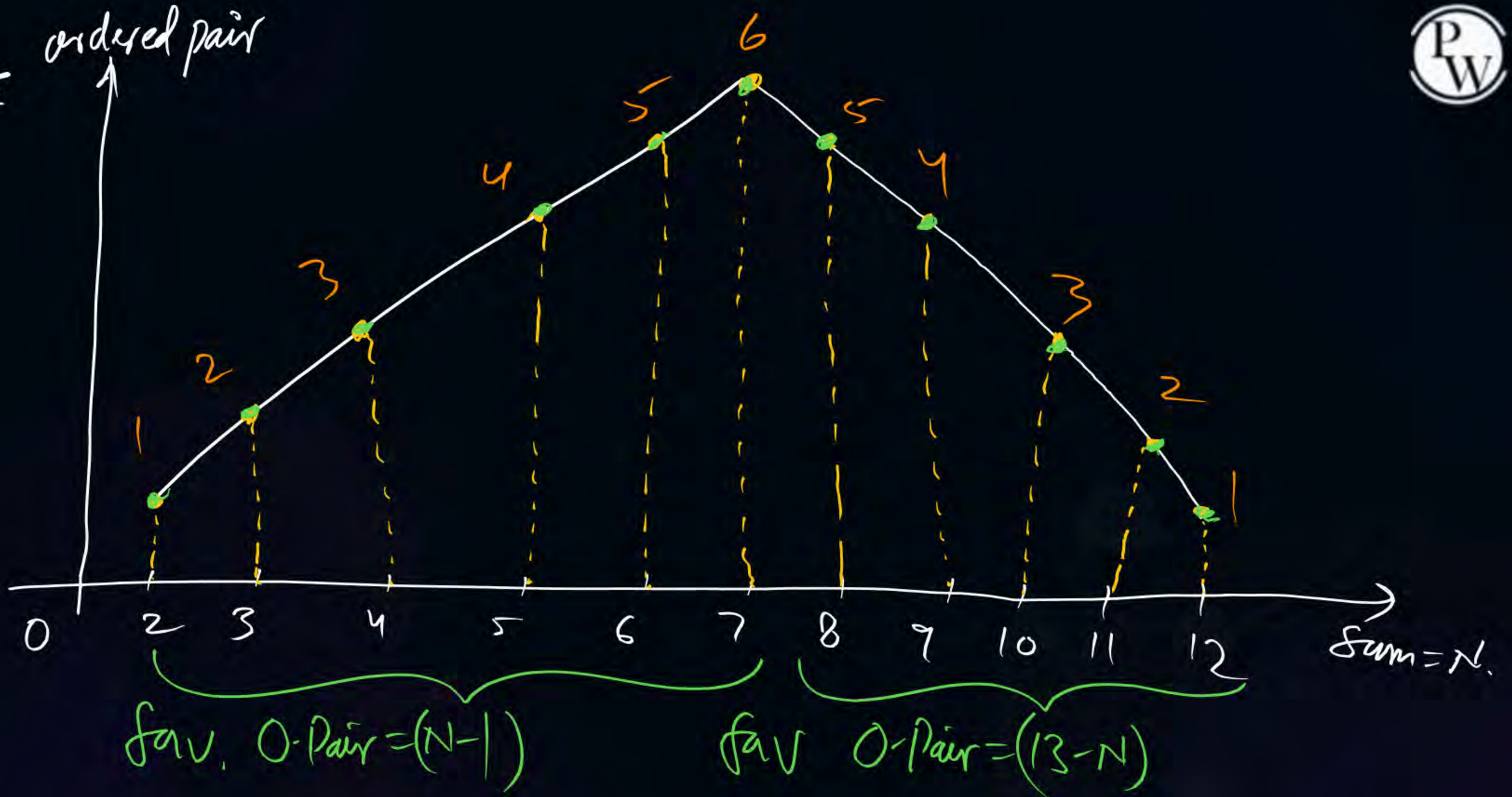
Abb I: Do yourself.

Trick: $P(\text{sum is divisible by } 4) = P(\text{sum} = 4 \text{ or } 8 \text{ or } 12) = \frac{3+5+1}{36} = \frac{9}{36}$

Sp Note: Individual elements of S. space are M.E at a time

$D = \{1, 2, 3, 4, 5, 6\}$, $D = \{(11)(12)(13)__(16), (21)____(66)\}$

Shortcut :-



fav. O-Pair = (N-1)

fav O-Pair = (13-N)

Qc Four Dice are thrown simultaneously & $P(\text{sum is } 22) = \frac{X}{1296}$ then $X = ?$

Sol: (App II) Total Cases $= \underline{6} \times \underline{6} \times \underline{6} \times \underline{6} = 6^4 = 1296$ Quadruples

Fav Quadruples $= \{$ eg (6664) .... $= \frac{4!}{3!} = 4$

or eg (6655) .... $= \frac{4!}{2!2!} = 6$

i.e fav outcomes $= 10$ so $P(\text{sum is } 22) = \frac{10}{1296} = \frac{X}{1296}$

$\Rightarrow X = 10$

PQ 2: A coin is tossed 6 times then write it's S.Space.

$$S=\{(HHHHHH)^{or},(HHHHHT)^{or},(HHHHTT)^{or},(HHHTTT)^{or},(HHTTTT)^{or},(HTTTTT)^{or},(TTTTTT)\}$$

$${}^6C_6=1 \quad {}^6C_5=6 \quad {}^6C_4=15 \quad {}^6C_3=20 \quad {}^6C_2=15 \quad {}^6C_1=6 \quad {}^6C_0=1$$

$$n(S)=\underset{C_1}{2}\times\underset{C_2}{2}\times\underset{C_3}{2}\times\underset{C_4}{2}\times\underset{C_5}{2}\times\underset{C_6}{2}=2^6=64 \text{ ordered 6 in tuples.}$$

$(HHHHHH)\longrightarrow \frac{6!}{6!}=1={}^6C_6$

$(HHHHHT)\longrightarrow \frac{6!}{5!}=6={}^6C_5$

$(HHHHTT)\longrightarrow \frac{6!}{4!2!}=15={}^6C_4$

$(HHHTTT)\longrightarrow \frac{6!}{3!3!}=20={}^6C_3$

$(HHTTTT)\longrightarrow \frac{6!}{2!4!}=15={}^6C_2$

$(HTTTTT)\longrightarrow \frac{6!}{5!}=6={}^6C_1$

$(TTTTTT)\longrightarrow \frac{6!}{6!}=1={}^6C_0$

In above E+P ; $P(H)=P(T)=\frac{1}{2}$ & Individual tosses are Independent

① Find the prob that all the outcomes are identical?

App I: fav elements $=\{(HHHHHH),(TTTTTT)\}$

$= 2$

So Req Prob $= \frac{fav}{Total} = \frac{2}{64} = \frac{1}{32}$

App II: Req Prob $= P[(HHHHHH) \text{ or } (TTTTTT)]$ → ME ←

$=\left(\frac{1}{2}\right)^6 + \left(\frac{1}{2}\right)^6 = \left(\frac{1}{2}\right)^5$

App III → Prob $= \frac{fav}{T} = \frac{{}^6C_6 + {}^6C_6}{2^6} = \frac{2}{64}$

② Find the prob that H & T appears alternately?

App I: fav Cases $=\{(HTHTHT),(THTHTH)\}$

$= 2$

Req Prob $= \frac{fav}{Total} = \frac{2}{64} = \frac{1}{32}$

App III: Req Prob $= P[(HTHTHT) \text{ or } (THTHTH)]$ → ME ←

$=\left(\frac{1}{2}\right)^6 + \left(\frac{1}{2}\right)^6 = \left(\frac{1}{2}\right)^5 = \frac{1}{32}$

(5) Find the prob that Both H & T appears at least once ?

App I: unfav outcomes $= \{(HHHHHH), (TTTTTT)\}$
$= 2$

So fav outcomes $= 64 - 2 = 62$

Req Prob $= \frac{\text{fav}}{\text{Total}} = \frac{62}{64}$

App III: Req Prob $= 1 - P(\text{all H or all T})$
$= 1 - \frac{2}{64} = \frac{62}{64}$

(6) Find the prob that H appears at least once ?

App I: unfav outcomes $= \{(TTTTTT)\} = 1$

So fav outcome $= 64 - 1 = 63$

Req Prob $= \frac{\text{fav}}{\text{Total}} = \frac{63}{64}$

App II: $P(\text{at least one H}) = 1 - P(\text{No Head})$
$= 1 - P(\text{all T})$
$= 1 - P(TTTTTT)$
$= 1 - \left(\frac{1}{2}\right)^6$

⑤ If 1st 2 outcomes are H, H then find the prob of occurring T when coin is tossed again

Sol:

Method-I

Req Prob = P(T in 3rd toss) = $\frac{1}{2}$

Method-II Approach III

∴

Req Prob = P[(HH) T something occurs]

(HH: given)

= 1 × 1 × $\frac{1}{2}$ × 1 × 1 × 1 = $\frac{1}{2}$ ≠ $\frac{32}{64}$

⑥ If 1st two outcomes are Both Heads then find the prob of occurring Tail in Remaining tosses ?

Approach III

Req Prob = P[(HH) TTTT]

(HH: given)

= 1 × 1 × $\frac{1}{2}$ × $\frac{1}{2}$ × $\frac{1}{2}$ × $\frac{1}{2}$

= $(1)^2 \times (\frac{1}{2})^4 = \frac{1}{16} \neq \frac{4}{64}$

Note:- Whenever Question is Based on Conditional Prob, we should avoid to generalise our answer.

i.e $\frac{32}{64} = \frac{1}{2}$ (✓) But $\frac{1}{2} = \frac{32}{64}$ ✗

(1) Find the prob that exactly 2 H appears ?

App I → fav Cases = {(HHTTTT), (HTHTTT), (TTTHTH), ...} ☹

= Not easy to count

App II → fav Cases = {eg (HTTHTT) .....}

$= \frac{6!}{2!\,4!} = {}^6C_2 = 15$

$\text{Req Prob} = \frac{\text{fav}}{\text{Total}} = \frac{{}^6C_2}{2^6} = \frac{15}{64}$

App III → use Binomial Dist → will be discussed later

(2) Find the prob that H & T appears equal Number of times ?

App I → No easy

App II → fav Cases = {H & T appears equal No of times}

= {exactly 3 H appears}

= {eg (HTHTTH) ....} $= {}^6C_3$

$\text{Req Prob} = \frac{\text{fav}}{\text{Total}} = \frac{{}^6C_3}{2^6} = \frac{20}{64}$

App III → using R-Distribution

⑨ Find the prob that, 1st two outcomes are Heads

Sol: App I fav cases = {(HH _ _ _ _ _ _) .....} = ?

App II fav cases = {eg (HH H/T H/T H/T H/T H/T .....}
(3rd, 4th, 5th, 6th)

$= 1 \times 1 \times 2 \times 2 \times 2 \times 2 = 16$

So Req Prob $= \frac{\text{fav}}{\text{Total}} = \frac{16}{64} = \frac{1}{4}$

App III Req Prob $= P[(HH \text{ Something occurs})]$

$= \frac{1}{2} \times \frac{1}{2} \times 1 \times 1 \times 1 \times 1 = \frac{1}{4}$

⑩ Find the prob that (only) 1st two outcomes are Heads)

App I fav cases = {(HH TTTT)} = 1

Req Prob $= \frac{\text{fav}}{\text{Total}} = \frac{1}{64}$

App II → No Need

App III

Req Prob $= P[\text{only 1st two Head}] = P[HH\,TTTT]$

$= \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2}$

$= \left(\frac{1}{2}\right)^2 \times \left(\frac{1}{2}\right)^4 = \left(\frac{1}{2}\right)^6$

Explanation of 89 →

$$\text{Req Prob} = \frac{\text{fav}}{\text{Total}} = \frac{(HH, HTHT, HTT, THHT, THTT, TTHH)}{\text{Total}}$$

$$= \frac{1\times1\times2\times2\times2\times2}{64}$$

App II

$$= \frac{1}{2}\times\frac{1}{2}\times\frac{2}{2}\times\frac{2}{2}\times\frac{2}{2}\times\frac{2}{2}$$

$$= \frac{1}{2}\times\frac{1}{2}\times1\times1\times1\times1 \Rightarrow P(HH \text{ somethg occurs})$$

= App III

$$64 = 2^6 = 2\times2\times2\times2\times2\times2$$

Diff b/n Q6 & Q10 -

(6) If 1st two outcomes are H, H then find the prob of occurring Tail in Remaining tosses

Condition

(App III) Req Prob = $P[\underset{\text{fix}}{(H\ H)}\ T\ T\ T\ T] = 1 \times 1 \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{16}$

(10) Find the Prob that only 1st two tosses produces Head

(App III) Req Prob = $P[H\ H\ T\ T\ T\ T] = \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \left(\frac{1}{2}\right)^6 = \frac{1}{64}$

HW8 A Coin is tossed 10 times then Find the prob that

① Exactly 4 H appears = ? $= \frac{\text{fav}}{\text{Total}} = \frac{{}^{10}C_4}{2^{10}} = \frac{\frac{10!}{4!6!}}{1024} = \frac{210}{1024}$

② 4th Head appears in 9th toss = ?

Ans: $\frac{56}{512}$

App II —

App III —

THANK - YOU